A.A. 2024-25. Exercise 1: Modelling of dynamic response of a R.C. structure in the elastic and inelastic range

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ABSTRACT: In this project, the modelling of an existing structure will be realised, using the software Midas. The models will be refined using different modelling methods. The results coming from a modal analysis, high intensity vibration dynamic analyses and a pushover analysis will be compared to experimental data coming from tests conducted on the structure in the laboratory. Finally, the model with the highest precision with respect to the real structure will be highlighted.

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1. DESCRIPTION OF THE STRUCTURE

1.1. Geometry

The structure under examination is the one shown in Figure 1 and 2. It is a dual system consisting of a frame with four columns and a coupled wall. The walls can have two different configurations: a rectangular shape and an L-shaped section.

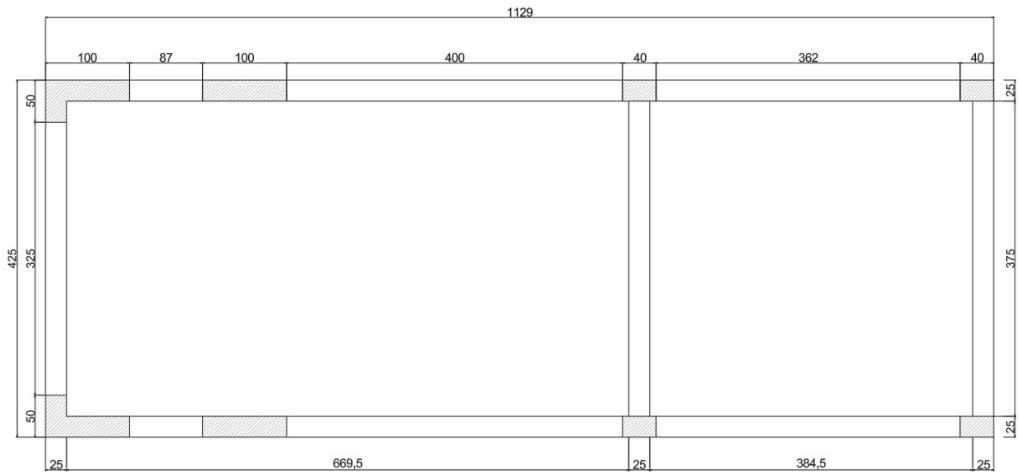

Figure 1: Plan view of the structure

The building has four stories, with a total height of 12.5 m, and transfers the loads to the foundation, which is 0.8 m height.

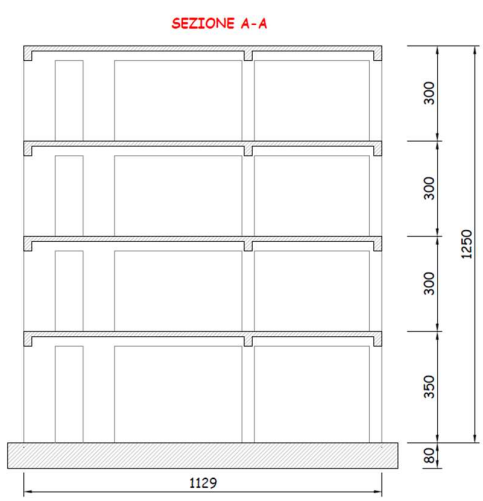

Figure 2: Section

The entire structure was designed to withstand a unidirectional earthquake in the test direction. However, based on the adopted detailing rules, it can be divided into two parts: one designed according to Eurocode 8 (EC8) and the other following the Displacement-Based Design (DBD) approach. The EC8 side was designed for a peak ground acceleration (PGA) of 0.4g, with soil class B (medium-dense sands or medium-stiff clays) and high ductility class (DCH).

This design approach was chosen to highlight the differences in reinforcement between the two sides, particularly the significant savings achieved with the DBD method, without notable differences in the overall seismic performance of the structure.

1.2. Displacement-Based Seismic Design

In displacement-based design approaches, seismic displacements are the primary parameter for structural design. Unlike Eurocode 8 (EC8), where forces dictate the acceptance criteria, in this method, all criteria are defined in terms of displacements.
The Displacement-Based Design (DBD) procedure can be summarized in four key steps:

1) Member Proportioning for Gravity Loads

In this phase, the structural elements are dimensioned considering both longitudinal and transverse reinforcement based on Ultimate Limit State (ULS) and Serviceability Limit State (SLS) design in flexure and shear. This includes capacity design of columns and ensuring minimum reinforcement requirements. Unlike conventional seismic design of reinforced concrete structures, where ductility requirements are imposed by seismic codes, DBD replaces these with detailing rules used for non-seismic structures.

2) Estimation of Inelastic Deformation Demands

The peak inelastic deformations of structural members are estimated under the design earthquake (corresponding to the 'life-safety' performance level). This step includes the calculation of inelastic chord rotation demands at member ends, typically considering 5% damping.

3) Reinforcement Adjustment and Final Detailing

The reinforcement layout and detailing of structural members are revised to ensure that their chord rotation capacity exceeds the seismic demands determined in the previous step.

4) Capacity Design for Shear and Structural Integrity

Shear capacity design is applied to both members and joints, ensuring that the revised reinforcement scheme satisfies shear strength requirements. Additionally, the capacity design of columns and all other members is re-evaluated for shear performance.

1.3. Reinforcement

The two sides of the structure were designed using different methodologies: the North side follows the Force-Based Design (FBD) approach, complying with Eurocode 8 for a high ductility class (DCH). In contrast, the South side adopts the Displacement-Based Design (DBD) methodology proposed by Panagiotakos and Fardis.

As previously mentioned, the DBD side requires approximately one-third less steel than the EC8-designed side across all three types of structural elements—walls, columns, and beams. This difference is evident in Figure 3, which illustrates the beam reinforcement, and Figure 4, which presents the reinforcement details for columns and shear walls.

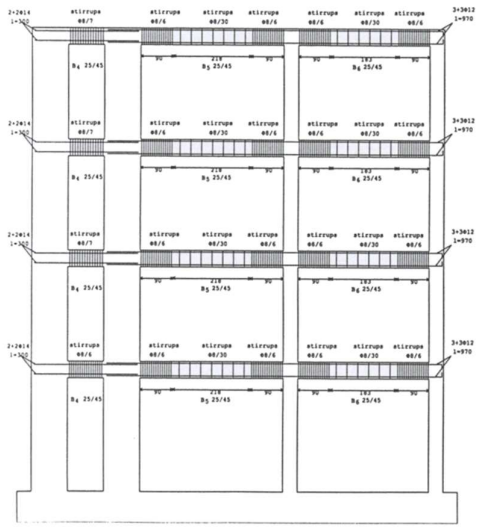

Figure 3: Reinforcement details of beams

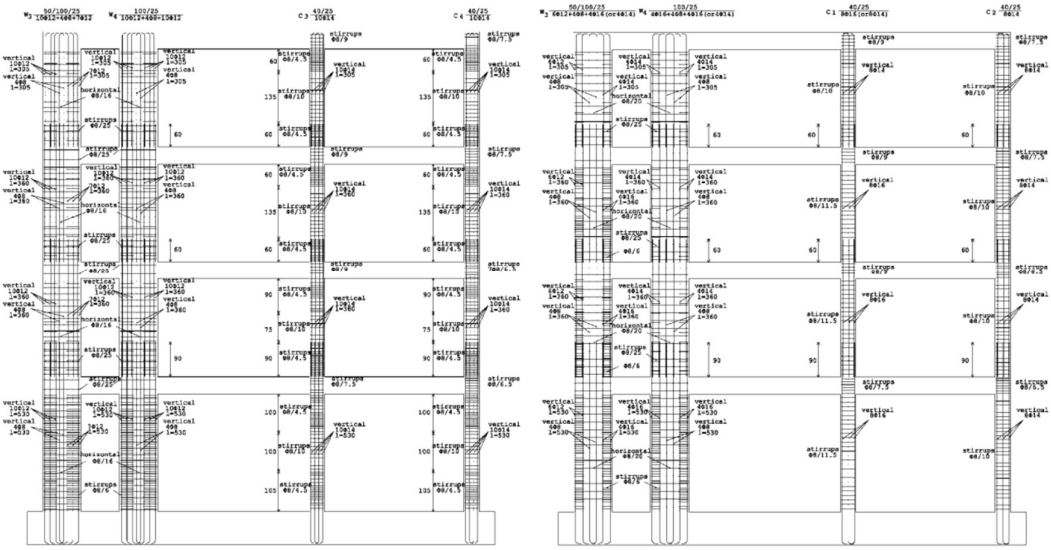

Figure 4: Reinforcement details of columns

1.4. Materials

The materials selected for the analyzed structure include concrete with a cylindrical compressive strength of 25MPa and high-ductility steel with a characteristic yield strength of 500MPa. Both materials underwent testing before and after the PsD tests, with the results presented in Table 1 for concrete and Table 2 for steel.

Table 1: Concrete Cylindrical Strength (MPa)

Location	Floor	28-day cubes	7-month cores	12-month cores
Columns	1	22.5		
Columns	2	21.2		
Columns	3	19.7		36.5 (1 core)
Columns	4	24.0		
Slab, beams	1	22.2	29.3	32.4 (3 cores)
Slab, beams	2	13.7	31.3	31.2 (3 cores)
Slab, beams	3	19.8	31.3	34.0 (2 cores)
Slab, beams	4	18.5	31.9	
Walls		22.4		
Walls		20.1		
Average		20.5	31.0	32.8

Table 2: Strength and ultimate elongation of steel bars

Diameter mm	Yield Strength MPa	Tensile Strength MPa	Ultimate elongation A5 %
8	557.5	642.1	23.7
10	525.2	617.2	24.2
12	516.5	615.0	24.8
14	521.9	615.2	26.2
16	506.4	627.0	24.0

2. MODAL ANALYSIS FOR LOW-INTENSITY VIBRATIONS

The initial phase of the structural study involves creating a detailed model of the structure to conduct modal and frequency analyses. The primary objective of this step is to minimize the relative error between experimental data and simulation results by refining and optimizing the model.

2.1. Experimental test

The Modal testing involves conducting vibration tests to identify the natural (modal) frequencies, modal masses, and mode shapes of the structure being analyzed. For the case under study, the experimental results were obtained through tests conducted at the ELSA laboratory and are presented in Table 3.

Table 3: Elsa Laboratory results for frequencies

Mode -	Frequency Hz	Period s	Type -
1	2.06	0.485	Direction Y
2	3.94	0.254	Torsion
3	4.31	0.232	Direction X
4	6.37	0.157	Direction Y
5	11.62	0.086	Direction Y
6	12.9	0.078	Torsion
7	15.32	0.065	Direction X
8	16.56	0.060	Direction Y
9	24.31	0.041	Torsion
10	31.5	0.032	Direction X
11	34.6	0.029	Torsion
12	45.87	0.022	Direction X

2.2. Structural Modeling

The structure, which experiences minor oscillations due to low-intensity vibrations, was modeled using the finite element software Midas Gen. An elastic model was developed, excluding considerations of concrete plasticity and cracking. Consequently, the configuration of longitudinal and transverse reinforcements, as outlined in section 1.3., was not incorporated into the model.

The model was constructed through the following steps:

1. Definition of Material Properties: The properties of C25/30 concrete were defined, including Poisson's ratio, elastic modulus, thermal coefficient, and specific weight.
2. Definition of Section Properties: The cross-sectional types of the structural elements, as described in section 1.2.1, were specified.
3. Definition of Model Geometry: The structural elements were modeled as "beam" elements in the software, adhering to the geometry and dimensions detailed in section 1.1.
4. Foundation Modeling: The foundation was approximated as a perfectly fixed support, reflecting its high stiffness.
5. Slab Modeling: Initially, the slab was modeled assuming an “Infinitely Rigid Diaphragm”, which reduced the structure's degrees of freedom to three per floor. This assumption implies that the horizontal diaphragm remains rigid within its plane, creating a mutual constraint among the structural nodes. Under this hypothesis, the nodes do not alter their relative distances in the X-Y plane and experience identical rotations around the Z-axis.

The displacements of the diaphragm nodes in the X-Y plane were linked to those of an auxiliary "Master" node, positioned at the center of mass of each diaphragm. This Master node was assigned translational mass and rotational inertia based on the dimensions and distribution of the structural elements.

Subsequently, the rigid diaphragm assumption was removed by modeling the slab using a discretized "plate" element. This step aimed to evaluate the impact of the rigid diaphragm hypothesis on the model's frequency outcomes.

The accompanying figure illustrates the Midas model of the structure, highlighting the masses assigned to the Master nodes and the perfectly fixed supports associated with the foundation slab.

2.3. Model 0: starting point

The initial model is the simplest one, intended to provide a general understanding of the structural behavior while offering a first approximation of the natural periods. To simplify the analysis, several assumptions have been made. Since the study is conducted within the elastic field, no reinforcement has been included in the model. The rigid diaphragm hypothesis is considered reasonable, as in-plane slab deformations are negligible compared to horizontal displacements. Moreover, the foundations are not explicitly modeled, with the columns assumed to be clamped to the ground. The influence of axial deformations in both columns and beams on the overall structural response has also been disregarded. In terms of mass distribution, translational masses and rotational inertias are assigned at the center of each floor based on the dimensions and arrangement of the structural elements.

The following table presents the weight values of the structural elements along with the calculation methods. For vertical loads, the sum of all elements on each floor has been considered, and the same approach has been applied to horizontal elements.

Table 4: Vertical Elements Weights

Element	Section m²	Mass density kg/m³	Linear mass kg/m
L-shaped column (100×50×25)	0.3125	2500	781.25
Walls (100×25)	0.25	2500	625
Columns (40× 25)	0.1	2500	250
Total			3812.5

Table 5: Horizontal Elements Weights

Element	Section m²	Length m	Mass/density kg/m³	Mass kg
Beam	0.1125	33.83	2500	9515
Slab	0.5625	10.54	2500	14822
Total				24337

Thanks to these data, the obtained weights can be summed, and by dividing this value by the acceleration due to gravity, the Translational Mass corresponding to each floor is determined. As for the Rotational Inertia, it is calculated using the following formula:

$$I_{zz} = \frac{M}{12}\left(L_x^2 + L_y^2\right)$$

where M represents the Translational Mass of the floor, Lx is the dimension of the structure along the x-axis (including the outer edge of the beams), and Ly is the corresponding length in the orthogonal direction.

Below is the table summarizing the steps to determine the Translational Mass and Rotational Inertia for each floor, which together define the concentrated mass assigned to the geometric centroid of each floor and are subsequently input into the Midas Gen software.

Table 6: Translational Mass and Rotational Inertia

Floor	Length of vertical element m	Mass of vertical elements kg	Total mass kg	Rotational inertia kN/g.m2	Translational mass kN/g
1	2.8	10675	35012	424.6	35.0
2 and 3	2.55	9722	34058	413.0	34.1
4	1.28	4861	29198	354.1	29.2

After defining the model, the analysis is carried out considering 12 modes, in line with the experimental study. To assess the quality of the results, the relative error between the periods obtained with the model and those measured in the laboratory is calculated using the follow equation:

$$\varepsilon = \frac{F_m - F_l}{F_m} \cdot 100$$

Where F_m is the frequency of the experimental one and F_l is the model. This error is analyzed first for all 12 modes and then focusing on the first 7, which account for more than 85% of the participating mass.

Table 7: Model 0: Error, Model Frequency and Experimental one

Mode	F_m Hz	F_l Hz	ΔF Hz	ε %
1	2.06	1.7840	0.276	13.398
2	3.94	3.6680	0.272	6.904
3	4.31	3.6732	0.6368	14.775
4	6.37	5.7689	0.6011	9.436
5	11.62	10.7466	0.8734	7.516
6	12.9	12.5257	0.3743	2.902
7	15.32	13.3313	1.9887	12.981
8	16.56	16.1688	0.3912	2.362
9	24.31	25.2183	-0.9083	3.736
10	31.5	28.7093	2.7907	8.859
11	34.6	40.2591	-5.6591	16.356
12	45.87	46.7893	-0.9193	2.004

Obtaining a mean relative error equal to 8.45%, considering all the 12 Modes. The error for mode 11 is particularly high. This suggests that for the first vibration modes, which account for nearly all the participating mass, the model frequencies are lower than the experimental ones. Consequently, the model appears to be less rigid than the actual structure.

2.4. Model 1: beam end offsets

The first aspect to address when increasing the stiffness of the structure is the joint region at the intersection of columns (or shear walls) and beams. While assuming this entire region to be infinitely rigid would significantly enhance the flexural stiffness of the members, in reality, diagonal cracking and shear deformation occur during seismic events. Therefore, it is more appropriate to consider only a portion of this region as infinitely rigid. As stated in *Seismic Design of Reinforced Concrete and Masonry Buildings* by T. Paulay and M. J. N. Priestley, approximately 20% of the interstorey deflection is typically attributed to joint deformations. For this reason, the authors recommend not treating these regions as fully rigid, except in the case of joints between beams and wide columns or shear walls, where deformations are less significant.

The proposed guidelines for defining the proportions of rigid joints are illustrated in the figure below.

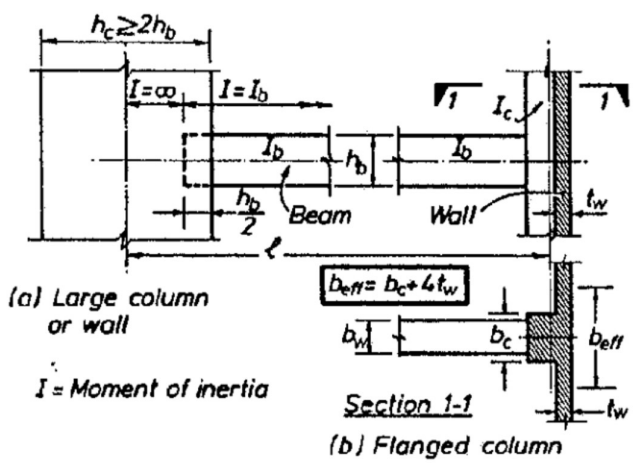

Figure 5 - Rigid joint dimensions

The software Midas automatically arranges the beams and columns converging at the beam-column joint so that the joint itself coincides with the centroid of all connected elements. Consequently, an offset was applied to the node position within the cross-sections of the elements, resulting in Model 1.

Table 8: Model 1: Error, Model Frequency and Experimental one

Mode	F_m	F_l	ΔF	ε
-	Hz	Hz	Hz	%
1	2.06	1.7740	0.286	13.883
2	3.94	3.4964	0.4436	11.259
3	4.31	3.5718	0.7382	17.128
4	6.37	5.7462	0.6238	9.793
5	11.62	10.7403	0.8797	7.571
6	12.9	12.1869	0.7131	5.528
7	15.32	12.9030	2.417	15.777
8	16.56	16.2468	0.3132	1.891
9	24.31	24.5197	-0.2097	0.863
10	31.5	28.2963	3.2037	10.170
11	34.6	39.1438	-4.5438	13.132
12	45.87	46.6084	-0.7384	1.610

Obtaining a mean relative error equal to 9.05%, considering all the 12 Modes. No improvement in error reduction was observed, as it increased in Model 1 for most modes. Additionally, frequency variations—and consequently stiffness changes—are not uniform across the different vibration modes. While the frequency generally decreases in most modes, an increase of stiffness is observed in mode eight.

2.5. Model 2: effective flange width

To make the model as close to reality as possible, it is necessary to consider the fact that the floor slab is solidly connected to the beams, increasing their stiffness. Specifically, slabs enhance the beams' resistance to rotation about their main axis.

For T and L beams, T. Paulay and M. J. N. Priestley recommend accounting for this aspect by considering the effective width of the section, as shown in the figure below.

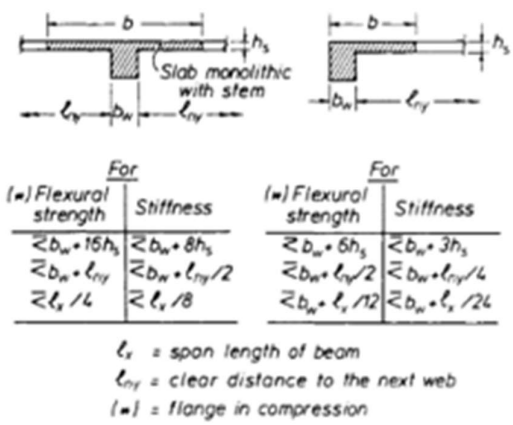

Figure 6: Assumptions foreffective width of beam flanges.

To compute the new effective width of each beam, the following numbering is used:

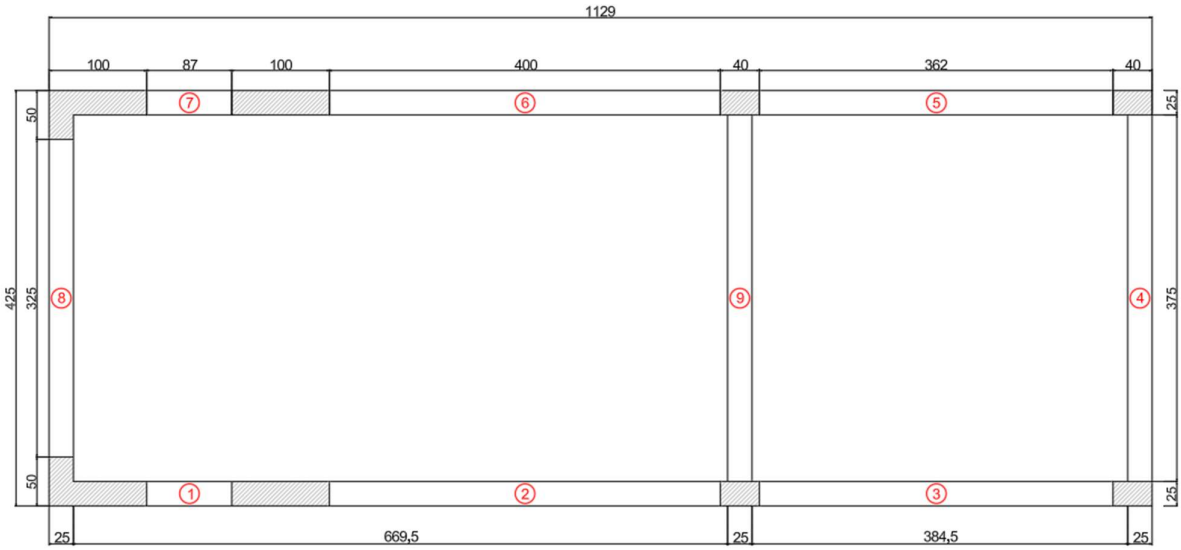

Figure 7: Beams' IDs in XY plane

With the cross-section of the beams 1-8 will become L-shaped profile, and the cross-section of the 9th beam modelled as a T-shaped profile. The computation of the effective width has been summarized in this table:

Table 9: Computation of b_{eff}

Beam number	b_w cm	l_x cm	h_s cm	l_{ny} cm	$b_w + k_1 h_s$ cm	$b_w + l_{ny}/k_2$ cm	$b_w + l_x/k_3$ cm	b_{eff} cm
1	25	224	15	375	70	118.75	34.33	34
2	25	470	15	375	70	118.75	44.58	45
3	25	410	15	375	70	118.75	42.08	42
4	25	400	15	384.5	70	121.125	41.67	42
5	25	410	15	375	70	118.75	42.08	42
6	25	470	15	375	70	118.75	44.58	45
7	25	224	15	375	70	118.75	34.33	34
8	25	400	15	694.5	70	198.625	41.67	42
9 (T-beam)	25	400	15	1054	145	552	50	50

After the computation, three different L-shaped sections and one T-shaped section have to be implemented in Midas. After this modification applied to all the beams in the structure, the eigenvalues analysis is carried out:

Table 10: Model 2: Error, Model Frequency and Experimental one

Mode	F_m	F_l	ΔF	ε
-	Hz	Hz	Hz	%
1	2.06	1.8104	0.2496	12.117
2	3.94	3.5892	0.3508	8.904
3	4.31	3.6913	0.6187	14.355
4	6,37	5.8365	0.5335	8.375
5	11.62	10.8448	0.7752	6.671
6	12.9	12.4417	0.4583	3.553
7	15.32	13.0161	2.3039	15.039
8	16.56	16.276	0.284	1.715
9	24.31	24.6965	-0.3865	1.590
10	31.5	28.2226	3.2774	10.404
11	34.6	39.1869	-4.5869	13.257
12	45.87	46.506	-0.636	1.387

Obtaining a mean relative error of 8.1138% across all 12 modes, a slight improvement in error reduction was observed. Additionally, frequency variations—and consequently stiffness changes—are not uniform across the different vibration modes. While the frequency generally increases in most modes.

2.6. Model 3: beam end offsets and effective flange width

Model 3 was created to bring the representation closer to the geometric reality of the structure by unifying the specific characteristics of models 1 and 2.

Table 11: Model 3: Error, Model Frequency and Experimental one

Mode	F_m	F_l	ΔF	ε
-	Hz	Hz	Hz	%
1	2.06	1.8120	0.248	12.039
2	3.94	3.6069	0.3331	8.454
3	4.31	3.6807	0.6293	14.601
4	6.37	5.8417	0.5283	8.294
5	11.62	10.8518	0.7682	6.611
6	12.9	12.4178	0.4822	3.738
7	15.32	13.0662	2.2538	14.711
8	16.56	16.2807	0.2793	1.687
9	24.31	24.6759	-0.3659	1.505
10	31.5	28.2753	3.2247	10.237
11	34.6	39.1800	-4.58	13.237
12	45.87	46.5338	-0.6638	1.447

Obtaining a mean relative error of 8.0467% across all 12 modes, this model shows a slight improvement over model 2. Specifically, it exhibits the same stiffness characteristics in the same modes.

2.7. Model 4: Adding of a plate

Model 4 has been built built from Model 1: the weight of the slabs was incorporated, enabling the system to compute the self-weights (eliminating the need for the concentrated masses used in previous models). This approach led to a gradual alignment of the model's frequencies with the experimental data, particularly as the slab's discretization was refined. A rectangular mesh was preferred over a triangular one, as it resulted in smaller discrepancies between the model's frequencies and the experimental values. The mesh size was kept sufficiently small to minimize errors in the analysis. Considering the natural period's definition, adjusting the mass distribution

can further enhance the results. As an initial step, to better approximate the real structure, the thickness of the plate is 15 cm.

Table 12: Model 4: Error, Model Frequency and Experimental one

Mode	F_m	F_l	ΔF	ε
-	Hz	Hz	Hz	%
1	2.06	1.8500	0.21	10.194
2	3.94	2.9905	0.9495	24.099
3	4.31	3.3985	0.9115	21.148
4	6.37	5.8894	0.4806	7.545
5	11.62	9.8993	1.7207	14.808
6	12.9	10.7190	2.181	16.907
7	15.32	12.2673	3.0527	19.926
8	16.56	15.4272	1.1328	6.841
9	24.31	18.7656	5.5444	22.807
10	31.5	19.0512	12.4488	39.520
11	34.6	22.2942	12.3058	35.566
12	45.87	22.9177	22.9523	50.038

Obtaining a mean relative error of 22.4499% across all 12 modes, this model shows a big error in the last 4 modes (from nine to twelve): the mean relative error is 15.1835%, still pretty high. The outcome shows a significant error, primarily caused by the under-stiffening of the structure.

2.8. Model 5: Panel zone effect

In Model 5, the panel zone effect was introduced to account for the additional rigidity in beam-column connections. In Midas Gen, this feature was activated using the *auto calculate panel zone effect* option. This command automatically detects the connections, calculates their stiffness based on beam and column dimensions, and applies a reduction factor. This feature ensures that the joint stiffness is properly considered, generating rigid connections where necessary. The panel zone effect cannot be applied to individual nodes but rather to the entire intersection region. The stiffness value can be adjusted from 0 (no stiffening) to 1 (fully stiffened intersection). After an iterative analysis, the results for the offset are:

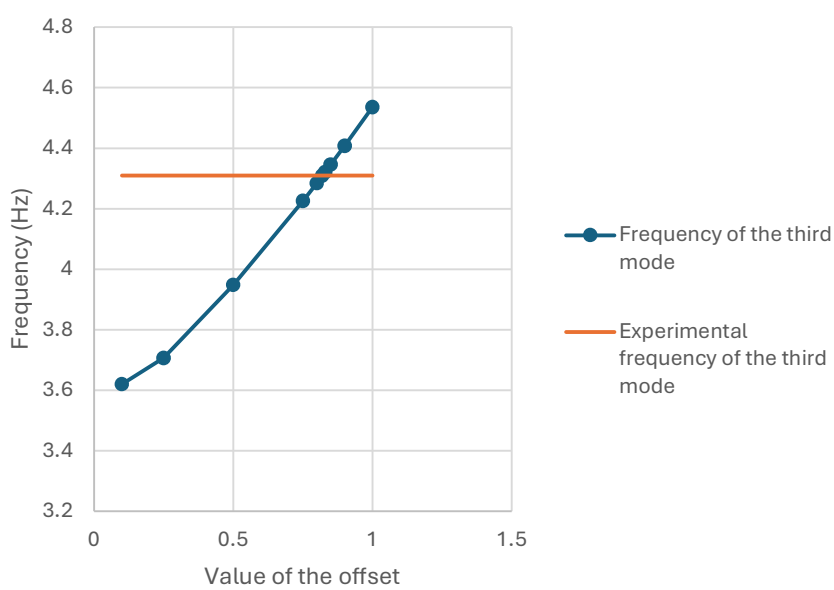

Figure 8: Iterative analysis performed to find the offset

Table 13: Model 5: Error, Model Frequency and Experimental one

Mode	F_m	F_l	ΔF	ε
-	Hz	Hz	Hz	%
1	2.06	1.8999	0.1601	7.771
2	3.94	4.0595	0.1195	3.032
3	4.31	4.3088	0.0012	0.027
4	6.37	6.1259	0.2441	3.832
5	11.62	11.4266	0.1934	1.664
6	12.9	13.5419	0.6419	4.975
7	15.32	15.2367	0.0833	0.544
8	16.56	17.149	0.589	3.556
9	24.31	26.4377	2.1277	8.752
10	31.5	31.9865	0.4865	1.544
11	34.6	41.0275	6.4275	18.576
12	45.87	50.9612	5.0912	11.099

Model 5, which includes the *panel zone effect*, shows a clear improvement in frequency accuracy compared to Model 1. In particular, the first six modes have an average error of 3.55%, with Mode 3 showing almost perfect agreement (0.03% error). This confirms that the additional stiffness at beam-column connections has been well accounted for in the lower frequencies.

For higher modes, however, errors increase significantly, reaching 18.58% in Mode 11 and 11.10% in Mode 12. This suggests that while the *panel zone effect* improves the model's stiffness representation, it may not fully capture the behavior of the structure at higher frequencies. Mode 9 also shows a notable discrepancy (8.75% error), reinforcing this trend.

The choice of a stiffness reduction factor of 0.82, determined through iterative analysis, appears effective in refining the response for low and mid-frequency ranges. However, the larger deviations at higher frequencies indicate that further refinements, such as a more detailed representation of joint behavior, may be necessary. Overall, considering the *panel zone effect* significantly improves the model, but some limitations remain for higher vibration modes.

2.9. Model 6: effective flange width & adding of a plate

The model 6 is made from the model 3 (T-shaped beams and offsets) with the addition of plate elements to model the floors.

Table 14: Model 6: Error, Model Frequency and Experimental one

Mode	F_m	F_l	ΔF	ε
-	Hz	Hz	Hz	%
1	2.06	1.81	0.219	10.631
2	3.94	3.08	0.865	21.942
3	4.31	3.49	0.812	18.835
4	6.37	5.84	0.534	8.375
5	11.62	10.0	1.581	13.603
6	12.9	10.5	2.355	18.258
7	15.32	12.3	3.012	19.664
8	16.56	14.99	1.5667	9.461
9	24.31	18.7	5.5722	22.921
10	31.5	19.7	11.8477	37.612
11	34.6	23.7	10.898	31.497
12	45.87	24.6	21.319	46.464

The average error is 21.6% for the twelve modes, which is much higher than the average error of model 3 (8.05%), which is the base for this model. It can be concluded that adding the plate model to the already present beams with effective flange width is not increasing the performance of the model.

2.10. Model 7: effective flange width, plate & panel zone effects

In this model, the model 6 is used with the added effect of the panel zone.

Table 15: Model 7: Error, Model Frequency and Experimental one

Mode	F_m	F_l	ΔF	ε
-	Hz	Hz	Hz	%
1	2.06	1.985	0.0747	3.626
2	3.94	3.368	0.5718	14.153
3	4.31	3.908	0.4025	9.339
4	6.37	6.314	0.052	0.882
5	11.62	10.93	0.6916	5.952
6	12.9	11.53	1.3715	10.632
7	15.32	13.54	1.7815	11.629
8	16.56	16.70	0.1427	0.862
9	24.31	20.37	3.94	16.207
10	31.5	20.67	10.831	34.383
11	34.6	24.80	9.7981	28.318
12	45.87	25.64	20.22	44.102

The mean error of the twelve first modes of this model is 15.04%, which is an improvement from model 6 (21.6%), but still higher than the reference model 3 (8.05%). Adding the panel zone effects improves the precision of the model in case of plates, but the results are further than the model without the plates.

2.11. Model 8: effective flange width & panel zone effect

In this model the slab collaboration was introduced combined with the offsets. For the perimetral beams, the width of collaboration is computed with the equation proposed by Paulay and Priestley:

$$b_{eff} = min\,\{b_w + l_x/24; b_w + 3h_s; b_w + l_{ny}/4\}$$
$$b_{eff} = b_w + 3h_s$$

Where b_w is the width of the beam and h_s is the height of the slab.

- No vertical offset provided
- A Panel zone effect of 0.75 provided

The numerical modal frequencies are compared with the experimental ones. An average error of 3.75% for the first 6 modes and 7.04% for the last 6 modes is obtained. The results are summarized on the table

Table 16: Model 8: Error, Model Frequency and Experimental one

Mode	Frequency	Experimental frequency	Modal frequency	Period	Absolute error	Average error
	rad/s	Hz	Hz	s	%	%
1	12.0155	2.06	1.9123	0.5229	7.17	
2	25.9156	3.94	4.1246	0.2424	4.69	
3	27.049	4.31	4.305	0.2323	0.12	
4	38.6974	6.37	6.1589	0.1624	3.31	
5	72.0188	11.62	11.4621	0.0872	1.36	
6	85.8133	12.9	13.6576	0.0732	5.87	3.75
7	94.8025	15.32	15.0883	0.0663	1.51	
8	107.4176	16.56	17.096	0.0585	3.24	
9	166.2733	24.31	26.4632	0.0378	8.86	
10	198.5785	31.5	31.6047	0.0316	0.33	
11	257.0939	34.6	40.9178	0.0244	8.25	
12	317.1875	45.87	50.482	0.0198	10.05	7.04

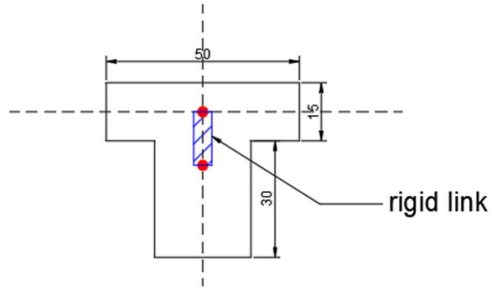

Figure 9: Rigid link generated in T-beam

The model generated a rigid link between the centroid of the T-beam and the slab reference axis. Which has resulted in increased stiffness. The increased stiffness increases the frequency and performance of the structure.

Finally, the most suitable models with respect to the modal analysis are model 5 and model 8. Model 5 gives better values for the first three modes, and especially for mode 3, which is the mode that will be the most excited during the next analysis. However, model 8 is showing very good results for the first three modes, and is more precise than model 5 for the higher modes. These two models will be used for the other steps of the analysis.

3. HIGH INTENSITY VIBRATIONS

Following the modal analysis, the structure underwent five additional tests with varying Peak Ground Acceleration, conducted through pseudo-dynamic tests. This hybrid method analytically models inertia forces and damping while experimentally capturing non-linear structural responses. A 0.005 sec time-step was used, subdividing calculations to iteratively determine displacements.

Table 17: Tests on the real structure

Test	Type	Design PGA amplification factor	PGA	Length
-	-	-	g	s
D01	Free oscillations			
D02	Pseudo Dynamic	0.05	0.028	8
D03	Pseudo Dynamic	0.10	0.56	15
D04	Pseudo Dynamic	0.15	0.085	4
D05	Pseudo Dynamic	1.00	0.560	15
D06	Pseudo Dynamic	1.50	0.850	6

For high-intensity excitations, linear analysis adjustments accounted for high-stress effects by incorporating reduced inertia for cracked sections and ductility factors for non-linearity. Additional masses simulated self-weight and dead loads, initially planned at 60 tons on the first floor, 59 tons on the second and third, and 54 tons on the roof. However, practical constraints led to differences in the applied loads, making precise axial load representation crucial for nonlinear analysis.

The vibration tests assumed a 10-meter tributary width per frame instead of 4 meters, with an additional 1.3 kN/m² to account for floor finishes, partitions, and live loads. This study focuses on the D05 case, representing the Damage Limitation requirement (PGA = 0.56g) per EC8. The input signal, derived from the 1995 Kobe earthquake, induced nonlinear behavior in the structure, leading to plastic hinge formation in critical sections. Displacements were constrained to the longitudinal direction, reflecting the predominant response of the system.

Midas Gen was used to input forces, later converted into masses. The nodal masses for linear elastic models accounted for self-weight and additional loads to align with experimental conditions. The results, including displacement time-histories and instantaneous fundamental frequencies, provided critical insights into structural behavior under extreme loads.

3.1. Modelling

The applied loads are depicted in the Table 18 and showed in Figure 10.

Table 18: Nodal loads applied, then converted into masses

Floor	L walls		Mid walls		Mid columns		Corner columns	
	Mass t	Load kN	Mass t	Load kN	Mass t	Load kN	Mass t	Load kN
Roof	3.78	37.08	8.10	79.46	9.72	95.35	5.40	52.97
3	4.13	40.52	8.85	86.82	10.62	104.18	5.90	57.88
2	4.13	40.52	8.85	86.82	10.62	104.18	5.90	57.88
1	4.20	41.20	9.00	88.29	10.80	105.95	6.00	58.86

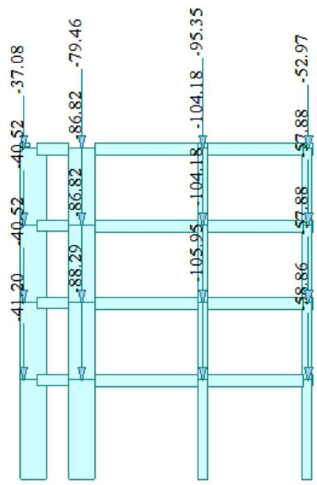

Figure 10: Nodal loads applied

For this analysis, we used the same models as those used for the eigenvalues analysis: model with offsets, model with offsets and panel zone effect, model with offsets and slab collaboration and model with slab collaboration and panel zone effect. The model with offsets and plate was not used because of the large difference in term of frequency. It is due to plates tending to act individually, decreasing the stiffness.

Additionally, cracking must be considered. Indeed, Figure 11 displays the evolution of the instantaneous fundamental frequency of the test structure. At 1.5s, it drops from a value of 2.7Hz which characterizes the uncracked structure to about 1.35Hz which shows the appearance of cracks.

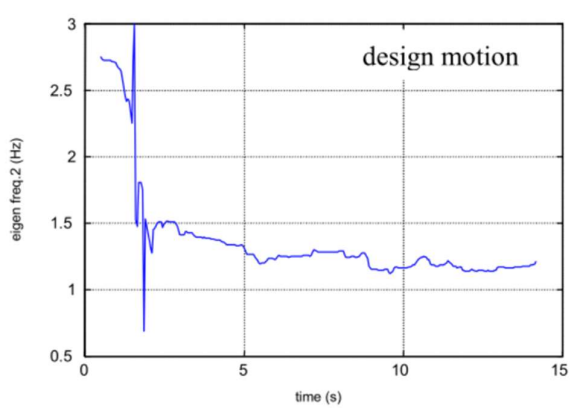

Figure 11: Evolution of the instantaneous fundamental frequency of the test structure

As a consequence, the stiffness of the members decreases. Figure 12 from FEMA 356 suggests values of stiffness for each component of the structure.

Chapter 6: Concrete

Table 6-5 Effective Stiffness Values

Component	Flexural Rigidity	Shear Rigidity	Axial Rigidity
Beams—nonprestressed	$0.5E_cI_g$	$0.4E_cA_w$	—
Beams—prestressed	E_cI_g	$0.4E_cA_w$	—
Columns with compression due to design gravity loads $\geq 0.5\ A_g f'_c$	$0.7E_cI_g$	$0.4E_cA_w$	E_cA_g
Columns with compression due to design gravity loads $\leq 0.3\ A_g f'_c$ or with tension	$0.5E_cI_g$	$0.4E_cA_w$	E_sA_s
Walls—uncracked (on inspection)	$0.8E_cI_g$	$0.4E_cA_w$	E_cA_g
Walls—cracked	$0.5E_cI_g$	$0.4E_cA_w$	E_cA_g
Flat Slabs—nonprestressed	See Section 6.5.4.2	$0.4E_cA_g$	—
Flat Slabs—prestressed	See Section 6.5.4.2	$0.4E_cA_g$	—

Note: It shall be permitted to take I_g for T-beams as twice the value of I_g of the web alone. Otherwise, I_g shall be based on the effective width as defined in Section 6.4.1.3. For columns with axial compression falling between the limits provided, linear interpolation shall be permitted. Alternatively, the more conservative effective stiffnesses shall be used.

Figure 12: Stiffness values suggested by FEMA 356

The structure is composed of non-prestressed beams and cracked walls. Concerning the columns, design gravity load must be compared to $0.3A_gf'_c = 750$kN and $0.7A_gf'_c = 1250$kN. All the values are inferior to $0.3A_gf'_c$. The factors can be directly applied to each model using "Section Stiffness Scale Factor".

Two types of analysis can be used to excite the structure: Time History and Response Spectrum.

3.2. Time History analysis

The first approach is the Time History. The structure is excited by a given ground acceleration.

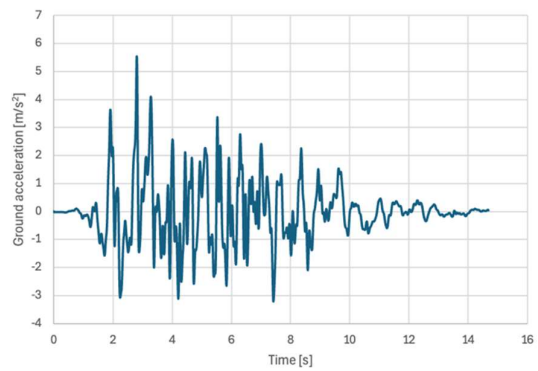

Figure 13: Accelerogram for the Time History analysis

3.2.1. Behaviour and ductility factors

In order to express the displacements obtained through the analysis, 2 factors are computed: the behaviour factor q and the ductility factor μ.

The q factor is taken equal to 5 and μ should be computed according to Eurocode 8 section 5.2.3.4(3) :

$$-If\ \mathrm{T}_1 \geq \mathrm{T}_C : \ \mu = q$$

$$-If\ \mathrm{T}_1 < \mathrm{T}_C : \ \mu = 1 + (q-1)\frac{T_c}{T_1}$$

The ductility factors were then computed for each model according to q=5 and their respective fundamental period.

Table 19: Behaviour factors q and ductility factors μ.

Model	q factor -	T s	μ -
Offsets	5	0.56	5.26
Offsets + panel zone effect	5	0.54	5.43
Offsets + slab collaboration	5	0.55	5.34
Offsets + slab collaboration + panel zone effect	5	0.52	5.59
RI + offsets	5	0.56	5.26
RI + offsets + panel zone effect	5	0.54	5.43
RI + offsets + slab collaboration	5	0.55	5.34
RI + offsets + slab collaboration + panel zone effect	5	0.52	5.59

3.2.2. Displacements

The displacements are computed with the following equation, using the elastic displacements obtained on Midas.

$$u_{max} = u_{el} \times \frac{\mu}{q}$$

The next figures show the obtained displacements compared to the experimental results.

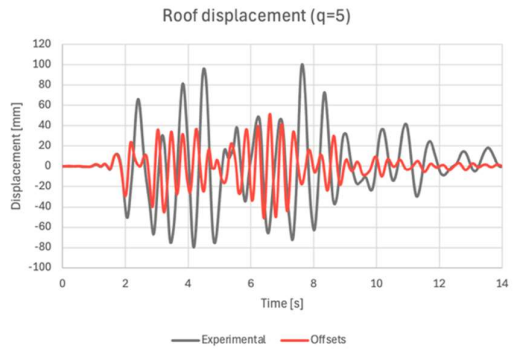

Figure 14: Roof displacement of the model with offsets

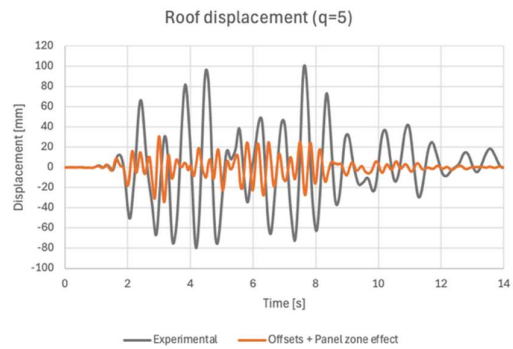

Figure 15: Roof displacement of the model with offsets and panel zone effect

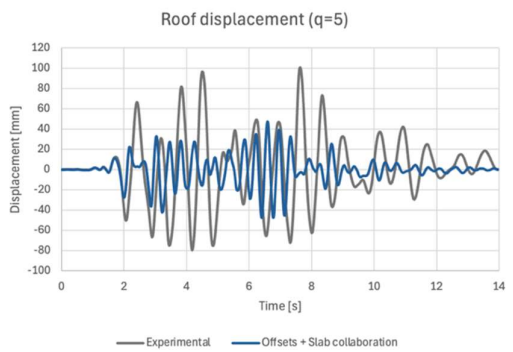

Figure 16: Roof displacement of the model with offsets and slab collaboration

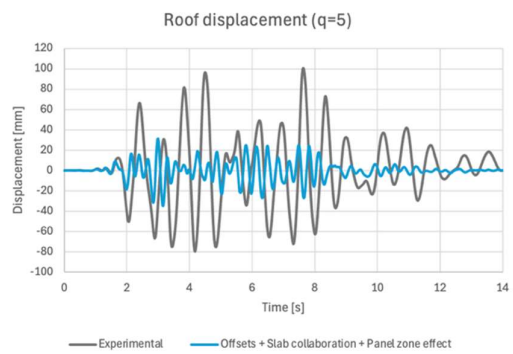

Figure 17: Roof displacement of the model with offsets, slab collaboration and panel zone effect

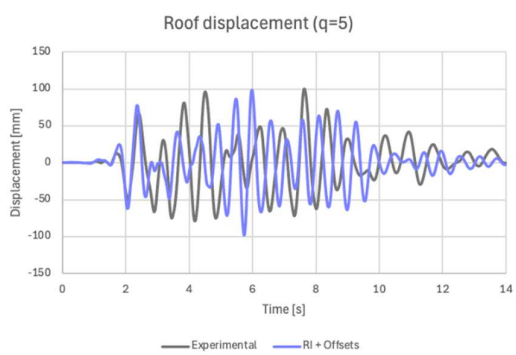

Figure 18: Roof displacement of the model with reduced inertia and offsets

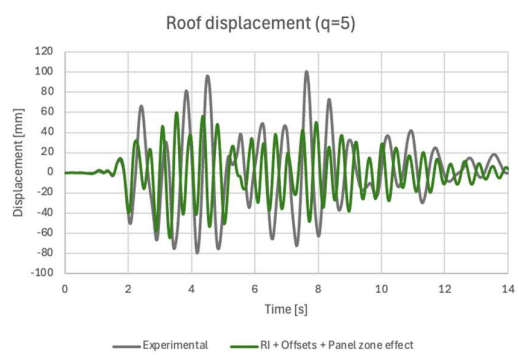

Figure 19: Roof displacement of the model with reduced inertia, offsets and panel zone effect

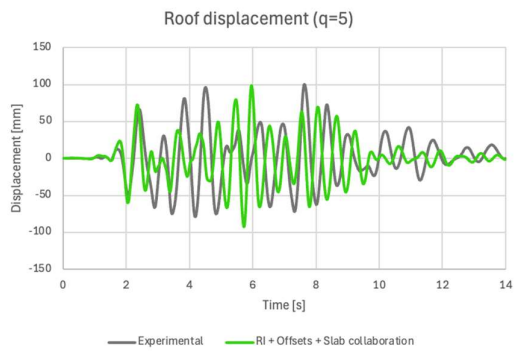

Figure 20: Roof displacement of the model with reduced inertia, offsets and slab collaboration

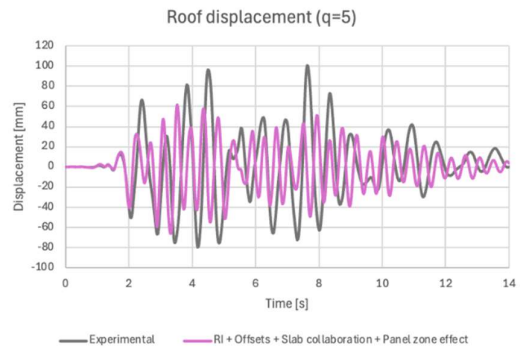

Figure 21: Roof displacement of the model with reduced inertia, offsets, slab collaboration and panel zone effect

The maximum displacements are displayed in the next table. The error with respect to the experimental maximum displacement is also computed.

Table 20: Maximum displacements and errors with q=5

Model	u_{max} mm	Error %
Experimental	100.69	
Offsets	51.59	48.76
Offsets + panel zone effect	30.80	69.41
Offsets + slab collaboration	47.79	52.54
Offsets + slab collaboration + panel zone effect	31.55	68.67
RI + offsets	98.21	2.46
RI + offsets + panel zone effect	60.39	40.02
RI + offsets + slab collaboration	98.39	2.28
RI + offsets + slab collaboration + panel zone effect	61.58	38.84

It can be osberved that the panel zone effect increases a lot the stiffness of the struture. Moreover, without reduced inertia, the displacements are very limited compared to the experimental ones.

When cracking is considered, by reducing inertia, the generated displacements are larger. Maximum displacements of models not considering panel zone effect reach a small error of about 2.35%.

3.2.3. Base shear

The base shear obtained with the time history analysis is larger than the experimental one. Indeed, according to Eurocode 8, it should be reduced by the q factor in order to take into account non-linearity and ductility.

$$V_{red} = \frac{V_{el}}{q}$$

Table 21: Base shear and reduced base shear

Model	Base shear kN	Reduced base shear kN
Experimental		1161.9
Offsets	1833.96	366.79
Offsets + panel zone effect	1935.43	387.09
Offsets + slab collaboration	1887.98	377.60
Offsets + slab collaboration + panel zone effect	1957.95	391.59
RI + offsets	1859.90	371.98
RI + offsets + panel zone effect	2123.70	424.74
RI + offsets + slab collaboration	1902.77	380.55
RI + offsets + slab collaboration + panel zone effect	2132.14	426.43

For every model, reduced base shear is significantly lower than experimental base shear. A new estimation of the behaviour factor can be computed by reversing the formula of reduced base shear in order to reach the experimental base shear. New ductility factors also have to be computed again, according to this new behaviour factor.

Table 22: New behaviour and ductility factors in order to reach experimental base shear

Model	New q factor -	T s	μ -
Offsets	1.58	0.56	1.62
Offsets + panel zone effect	1.67	0.54	1.74
Offsets + slab collaboration	1.62	0.55	1.68
Offsets + slab collaboration + panel zone effect	1.69	0.52	1.79
RI + offsets	1.60	0.56	1.62
RI + offsets + panel zone effect	1.83	0.54	1.74
RI + offsets + slab collaboration	1.64	0.55	1.68
RI + offsets + slab collaboration + panel zone effect	1.84	0.52	1.79

The obtained q factors are significantly lower.

Finally, displacements can be computed according to these new estimations of behaviour and ductility factors.

Table 23: Displacements according to new behaviour and ductility factors

Model	u_{max} mm	Error %
Experimental	100.69	
Offsets	50.22	50.12
Offsets + panel zone effect	29.57	70.63
Offsets + slab collaboration	46.19	54.13
Offsets + slab collaboration + panel zone effect	29.91	70.29
RI + offsets	95.66	4.99
RI + offsets + panel zone effect	58.30	42.10
RI + offsets + slab collaboration	95.14	5.52
RI + offsets + slab collaboration + panel zone effect	58.78	41.62

For each model, errors are quite similar to the ones with q=5.

3.3. Response Spectrum analysis

The second approach is Response Spectrum. Two elastic response spectra are used: the first one is the experimental one, specified in Eurocode 8 - 1994, and the second is the one specified in Eurocode 8 - 2004.

The spectra were generated using a PGA of 0.4g, soil type B and an elastic dumping factor of 5%.

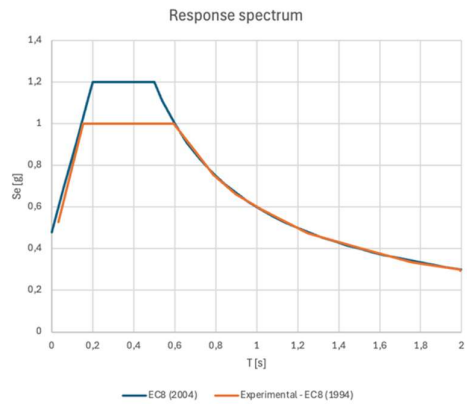

Figure 22: Elastic response spectra

The procedure is the same as for Time History, except that for EC8 – 2004, the ductility factor is equal to the behavior factor because $T_1 \geq T_c$.

3.3.1. Behaviour and ductility factors

Table 24: Behaviour and ductility factors for EC8 – 1994

Model	q -	T s	μ -
Offsets	5	0.56	5.26
Offsets + panel zone effect	5	0.54	5.43
Offsets + slab collaboration	5	0.55	5.34
Offsets + slab collaboration + panel zone effect	5	0.52	5.59
RI + offsets	5	0.56	5.26
RI + offsets + panel zone effect	5	0.54	5.43
RI + offsets + slab collaboration	5	0.55	5.34
RI + offsets + slab collaboration + panel zone effect	5	0.52	5.59

Table 25: Behaviour and ductility factors for EC8 – 2004

Model	New q factor -	T s	μ -
Offsets	5	0.56	5
Offsets + panel zone effect	5	0.54	5
Offsets + slab collaboration	5	0.55	5
Offsets + slab collaboration + panel zone effect	5	0.52	5
RI + offsets	5	0.56	5
RI + offsets + panel zone effect	5	0.54	5
RI + offsets + slab collaboration	5	0.55	5
RI + offsets + slab collaboration + panel zone effect	5	0.52	5

3.3.2. *Displacement*

Table 26: Maximum displacements and errors with q=5 for EC8 – 1994

Model	u_{max} mm	Error %
Experimental	100.69	
Offsets	50.81	49.54
Offsets + panel zone effect	34.33	65.90
Offsets + slab collaboration	48.74	51.60
Offsets + slab collaboration + panel zone effect	34.83	65.41
RI + offsets	99.83	0.85
RI + offsets + panel zone effect	64.05	36.39
RI + offsets + slab collaboration	95.71	4.95
RI + offsets + slab collaboration + panel zone effect	65.38	35.06

Table 27: Maximum displacements and errors with q=5 for EC8 – 2004

Model	u_{max} mm	Error %
Experimental	100.69	
Offsets	57.87	42.53
Offsets + panel zone effect	37.83	62.43
Offsets + slab collaboration	54.60	45.77
Offsets + slab collaboration + panel zone effect	37.31	62.94
RI + offsets	105.88	-5.15
RI + offsets + panel zone effect	70.57	29.91
RI + offsets + slab collaboration	102.54	-1.84
RI + offsets + slab collaboration + panel zone effect	70.04	30.44

3.3.3. *Base shear*

Table 28: Base shear and new behaviour and ductility factors for EC8 – 1994

Model	Base shear kN	Reduced base shear kN	New q factor -	T s	μ -
Experimental		1161.9			
Offsets	1875.00	375.00	1.61	0.56	1.65
Offsets + panel zone effect	1912.54	382.51	1.65	0.54	1.72
Offsets + slab collaboration	1904.18	380.84	1.64	0.55	1.69
Offsets + slab collaboration + panel zone effect	1938.62	387.72	1.67	0.52	1.77
RI + offsets	1890.68	378.14	1.63	0.56	1.67
RI + offsets + panel zone effect	1942.30	388.46	1.67	0.54	1.74
RI + offsets + slab collaboration	1919.50	383.90	1.65	0.55	1.71
RI + offsets + slab collaboration + panel zone effect	1966.68	393.34	1.69	0.52	1.79

Table 29: Base shear and new behaviour and ductility factors for EC8 – 2004

Model	Base shear kN	Reduced base shear kN	New q factor -	T s	μ -
Experimental		1161.9			
Offsets	2238.44	447.69	1.93	0.56	1.99
Offsets + panel zone effect	2285.32	457.06	1.97	0.54	2.07
Offsets + slab collaboration	2274.32	454.86	1.96	0.55	2.04
Offsets + slab collaboration + panel zone effect	2317.12	463.42	1.99	0.52	2.14
RI + offsets	2103.98	420.80	1.81	0.56	1.86
RI + offsets + panel zone effect	2319.94	463.99	2.00	0.54	2.10
RI + offsets + slab collaboration	2193.28	438.66	1.89	0.55	1.96
RI + offsets + slab collaboration + panel zone effect	2349.90	469.98	2.02	0.52	2.17

Table 30: Displacements according to new behaviour and ductility factors for EC8 – 1994

Model	u_{max} mm	Error %
Experimental	100.69	
Offsets	49.51	50.83
Offsets + panel zone effect	32.94	67.29
Offsets + slab collaboration	47.13	53.20
Offsets + slab collaboration + panel zone effect	33.00	67.23
RI + offsets	97.30	3.37
RI + offsets + panel zone effect	61.50	38.92
RI + offsets + slab collaboration	92.58	8.05
RI + offsets + slab collaboration + panel zone effect	62.01	38.41

Table 31: Displacements according to new behaviour and ductility factors for EC8 – 2004

Model	u_{max} mm	Error %
Experimental	100.69	
Offsets	59.66	40.75
Offsets + panel zone effect	39.85	60.42
Offsets + slab collaboration	56.90	43.49
Offsets + slab collaboration + panel zone effect	40.05	60.22
RI + offsets	108.83	-8.18
RI + offsets + panel zone effect	74.39	26.11
RI + offsets + slab collaboration	106.70	-5.97
RI + offsets + slab collaboration + panel zone effect	75.26	25.26

3.4. Comparisons and conclusions

In terms of displacement errors, for each model, similar values are obtained for Time History and both Response Spectra. It can be noticed that the models with the highest errors are the ones including panel zone effect. Moreover, the impact of the stiffness reduction to consider cracking can be clearly observed. Finally, the selection of the behaviour factor is crucial for the modelling of the structure.

4. PUSHOVER ANALYSIS

4.1. Model used and modifications

The models used for the pushover analysis are the model 5 and the model 8, with some modifications and added characteristics. Nodal forces are introduced to take into account the added masses present in the experiment. The floor diaphragm masses are replaced by these values of the nodal masses :

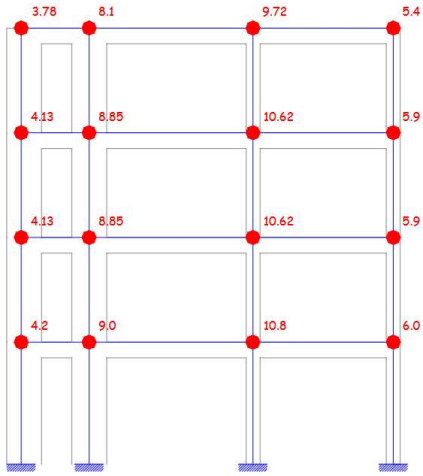

Figure 23: Nodal masses distribution

To introduce these values of the masses in Midas, the corresponding load is set up at each node as a nodal force, and is then converted to a mass through the Load to Masses function of Midas.

Non-Linearity only enters when the element yields, prior to that all elements are in yield limit which makes them linear and hence platic hinges are to be introduced to account for non-linearity. Non-linearity can be provided by either reducing the stiffness by FEMA provisions by decreasing the cross sectional properties or can be adopeted by manually providing the plastic hinge properties. In this we adopted the later case.

In this regard, the reinforcement has to be introduced to compute the plastic hinges of the model. For the columns and beams with rectangular shape and simple rebar layout, it is possible to define directly in Midas Gen their reinforcement in the Rebar Data window. A separate method has to be realized for the L-shaped and rectangular walls, in the case of model 5, and for the L and T-shaped beams in model 8. The reinforcement data has been taken from the reinforcement layout provided. One side of the building's reinforcement is designed in accordance with Eurocode 8, the other one with reference to DBD. For the beams, two separate cases have to be taken into account for both EC8 and DBD. Indeed for the EC8 side, beam reinforcement is different from the left part to the right part of the building, but is the same on all stories. For the DBD side, the difference is between the first interstory, which has more rebars compared to the three interstories above. For model 5, the beams are rectangular, and after defining the rebar layout for a considered element, the plastic hinge properties are computed automatically by Midas using the command Assign Pushover Hinge Properties.

For the L-shaped and rectangular walls in case of model 5 and 8, and for L and T-shaped beams in model 8, it is not possible to define their reinforcement directly in Midas Gen, because of their non regular shape and rebar layout. It is then necessary to use Midas GSD, which allows to define manually and precisely the position and diameter of each reinforcement. Midas GSD is used to compute the Moment-Curvature curve for each section considered. This curve is dependant of the axial load, which has to be imputed and is different for every level of the building. The value of the axial load, for each interstories and walls, is taken from a Midas Gen analysis. After imputing the value of the axial load, the Moment-Curvature curve is bilinearized, and two points from the

graph are extracted : the curve and moment corresponding to the yielding and to the ultimate strain. If the section is not symmetric, it is necessary to extract also these two points at an angle of 90°. These points are then imputed into Midas Gen for the definition of the Pushover Hinges, through the Define Pushover Hinge Properties function. After these steps, every plastic hinges in the structure are defined and the pushover analysis can proceed.

4.2. Pushover Analysis Parameters

Two separate pushover analysis are conducted: one with a constant mode shape, and another one with uniform acceleration.

4.3. Pushover Analysis results

In the pushover analysis, non-linearity is introduced by providing plastic hinge properties through Moment-Curvature curve by obtaining bi-linearied curve from Midas GSD and providing them manually in Midas Gen.

Hence we find these non-linear Capacity-curve for model 5 and model 8:

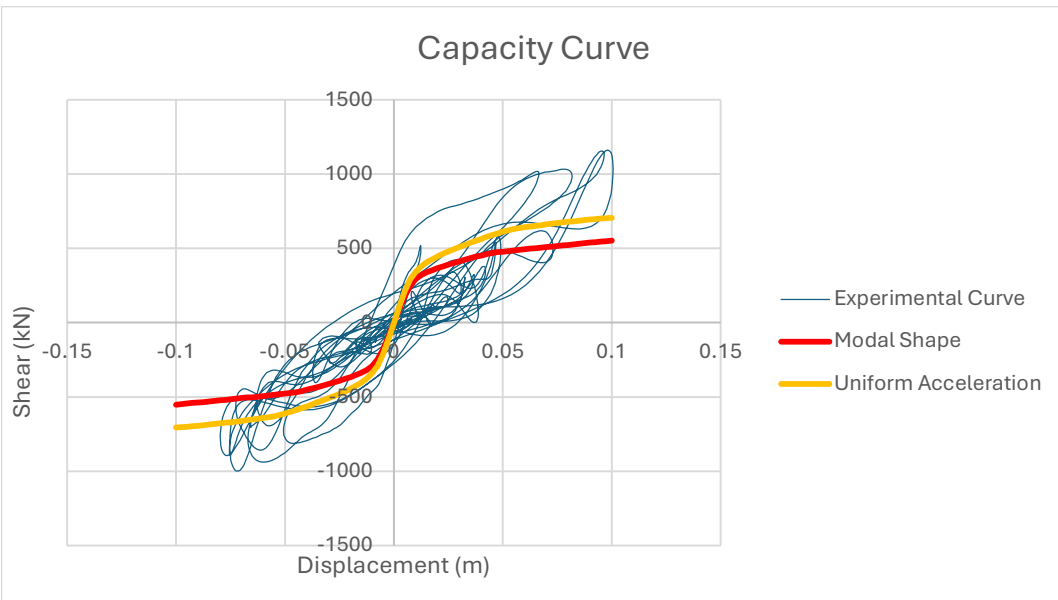

Figure 24: Model 5 capacity curve

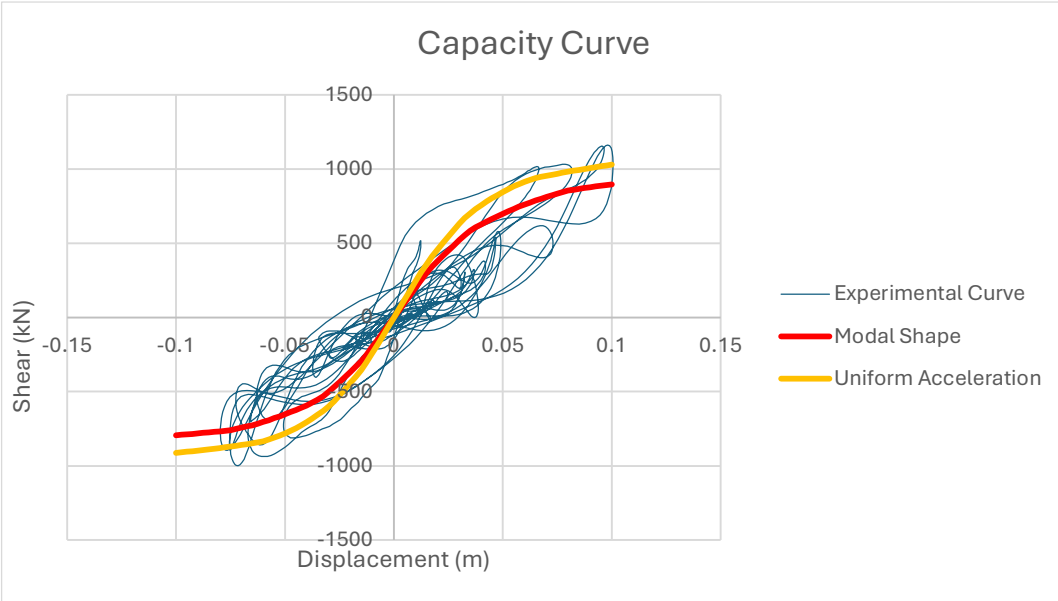

Figure 25: Model 8 capacity curve

It can be seen that for the given displacement control of 100mm, the curve obtained from push-over analysis with uniform acceleration of model 8 is closer to the experimental curve than model 5, with value of the shear reaching up to 1030kN, compared to 705kN for model 5, while the maximum shear reached in the experimental test is 1160 kN.

Table 32: Comparison between models 5 and 8

Model	Shape of beams	Frequency error of 1st 6 modes %	Formation of rigid links ?	Shear capacity for 100 mm displacement kN
5	Rectangular	3.55	No	705kN
8	T beams with slab collaboration	3.75	Yes	1030kN

Due to the shape of the beams, the beam stiffness is higher for model 8. Moreover, frequency error is inferior in model 5. Finally, formation of rigid links in Model 8 increases the stiffness and therefore the shear carrying capacity of Model 8. The model 8 is more suitable for the pushover analysis than model 5.

Model 8 is opted as there was no vertical offset of 15cm provided and there was rigid link formation between the centroid of the slab and the beam thereby increasing the stiffness of the system compared to the other models. The rigid link formation is explained under modal analysis results of model 8.

$$\omega = \sqrt{\frac{k}{m}}$$

$$\omega = 2\pi f$$

$$f \; \alpha \sqrt{k}$$

The frequency increases with increase in stiffness hence the difference of the frequency of different modes of model 8 was lesser than the experimental ones compared to other models considered. Which resulted in the better capacity curve implying the strength very close to the strength of the experimental model.

The experimental curve (hysteresis loop) indicates that the structure has passed the yield state. On loading and unloading due to cyclic loads the structure has reached plastic state and has started to dissipate energy and unloading curve doesn't closely follow the loading curve indicating major loss in energy. Since the path of the curve also has changed and not constant implies that there has been permanent deformation. Hence concluding that the structure has formed sufficient plastic hinges. Since there is no specification on collapse of the structure during the experimental test, it can still dissipate energy and form few more plastic hinges and hence ductility is not fully exploited.

5. CONCLUSION

The aim of this project was to model an existing structure and compare it with data from experimental tests. For the modal analysis, models were find with very high precision for the first modes, which are the most important to simulate the response of the structure under seismic excitation. These models were used in high intensity vibration dynamic analyses, and showed good results close to the experimental test. Finally, after introducing the plastic hinge characteristics, a pushover analysis was conducted to evaluate the non-linear behaviour of the models, which, again, is close to the real structure response.
It is important to be able to model precisely buildings and other structures, in order to test and evaluate their characteristics without using analysis on large scale specimen in the laboratory that are costly and time consumming. These models can be used to perform the design of the building, allowing the computation of the size of the elements composing the building and their reinforcement.

BIBLIOGRAPHY

Colombo, A. et al. 2001. *Pseudodynamic Response of Dual RC Structure with Different Designs.* Elsevier Science Ltd.

Martinelli, L. 2004. *Numerical Modeling of a PSD Test on a Dual RC System*. Proceedings of 13th World Conference on Earthquake Engineering. Vancouver, B.C, Canada. Paper no. 2156

Panagiotakos, T. B., and Fardis, M. N. 2001. A Displacement-Based Seismic Design Procedure for RC Buildings and Comparison with EC8. *Earthquake Engineering and Structural Dynamics.* 30(10):1439–1462.

Paulay T, Priestley NJM. *Seismic design of reinforced concrete and masonry buildings*. J Wiley&Sons. 1992.

Pegon, P., and Pinto, A. V. 2000. Pseudo-Dynamic Testing with Substructuring at the ELSA Laboratory. *Earthquake Engineering and Structural Dynamics.* 29:905–925.